

CREATING A MULTI-CAPILLARY FURNACE (MCF)

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Abstract

The Spectroscopy beamlines at Diamond Light Source regularly have operando/in-situ of X-ray Absorption Fine Structure (XAFS) experiments on catalyst samples. To improve the efficiency of such experiments, a Multi-capillary Furnace (MCF) has been designed to enable the measurement of multiple catalysts in parallel reactions. Subsequent testing has proven that the furnace can have up to four samples at 600 °C without thermal ‘cross-talk’ and permits samples to be remotely moved into the beam path.

INTRODUCTION

In the existing furnaces at the Spectroscopy beamlines at Diamond Light Source, powder samples are placed inside cylindrical 3mm diameter capillaries, before then being lowered into ceramic furnaces where they are heated for individual catalysis experiments. In these furnaces, only one sample can be loaded at a time. Consequently, replacing the capillaries/samples for the next experiment causes time to be lost during the changeover, reducing beamtime efficiency. The objective therefore was to create a furnace where up to four samples can be placed inside in separate capillary channels, where each sample could be heated and tested independently of each other, whilst allowing remote operation to move between channels to improve efficiency.

OBJECTIVES AND DELIVERABLES

The I18, I20 and B18 beamlines which make up the Spectroscopy group at Diamond Light Source have experimental tables which were already set up with limited space available for the mounting of a furnace. As a result, it was not simply a case of making four separate single capillary assemblies, as the space requirement would have been too large to fit them as opposed to a new Multi-capillary Furnace (MCF).

Catalysis experiments usually involve reactions of catalysts with gases and temperature/pressure. Some of these samples can require long pre-treatment (e.g. activation) before being measured, so having multiple capillaries with independent and stable temperature increase control allows reactions in one sample to continue while another is measured.

The MCF was to be tested and commissioned both offline (e.g. in a separate laboratory) and within the B18 beamline. To assess the suitability of the MCF, acceptance criteria were created to determine whether the MCF fulfils the required functions of simultaneous testing on several samples. The following key acceptance criteria was established as part of the furnace design:

- Temperature at the sample can be controlled from temperatures of at least 600 °C, and independently for each of the four capillaries.

- Temperature stability at the sample within ± 5 °C.
- Temperature ‘cross-talk’ between the samples within ± 5 °C.

FURNACE CONCEPT DESIGN AND FINITE ELEMENT ANALYSIS (FEA)

The MCF design concept uses kanthal wire heating elements housed in ceramic inside the furnace body as an individual ‘box-furnace’ unit. This unit is powered by a separate supply with independent temperature control. Individual samples are then located between the heating element and covered by additional ceramics (with apertures for the beam path) to aid heating samples to temperature.

A ‘single-cell’ prototype of the design (Fig. 1) was developed to determine if the sample could reach 600 °C as required with the proposed design, before being expanded to the MCF. To raise the temperature at the sample, the voltage from the power supply was steadily increased, and the temperature recorded using thermocouple probes placed within 10 mm of the sample position.

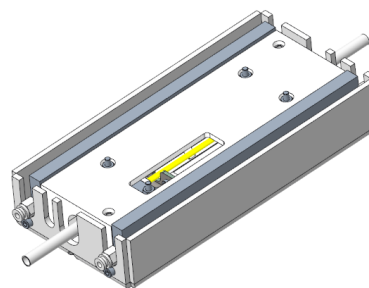
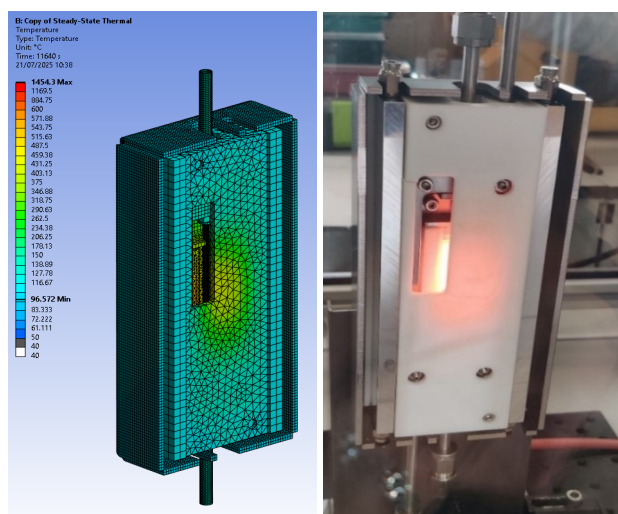


Figure 1: Prototype furnace ‘single cell’ with ‘box-furnace’ unit and capillary inside.

The body was made of sheet stainless steel, which allows for the mounting of the furnace units but also is less conductive of heat than other types of metal, with the theory being that the ceramics will contain the heat at the sample in order to keep it at 600 °C whilst ensuring the furnace body does not reach too high a temperature.

The prototype testing showed that the samples could reach temperature, but the ‘policeman’ thermocouples on the furnace body reached a temperature of ~ 200 °C (too high for safe user contact), and the ceramics cracked due to the temperature increase. Finite Element Analysis (FEA) in Ansys was used to represent the experiment conditions, and results showed large temperature peaks in areas where ceramics got hot and cracks propagated (Fig. 2).



Figures 2: FEA results showing temperature peaks around ceramics, and comparison to prototype.

DESIGN DEVELOPMENT AND PREPARATION FOR BEAMLINE

To solve the issue of the high body temperatures and cracking ceramics, a design iteration was necessary which would find a way to keep the outer body of the furnace cooler that did not prevent the samples from reaching the temperatures needed for the involved experiment. It was proposed that the body was changed to a larger aluminium block with integrated cooling channels using a raw water supply (Fig. 3). These channels not only aid in ensuring the whole furnace remains cool throughout the heating process, but also in thermally isolating the samples from each other to avoid temperature ‘cross-talk’.

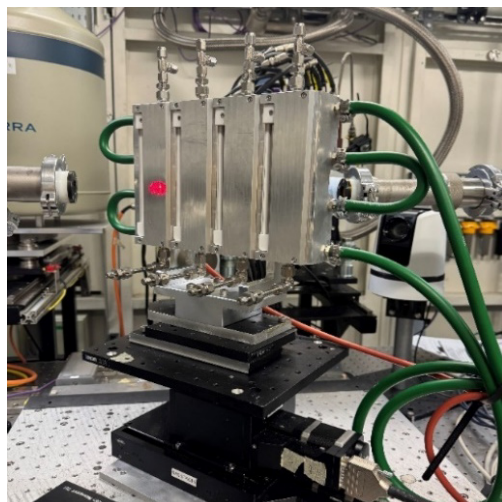


Figure 3: MCF in beamline with cooling channels.

FEA was used again to verify that both the body and ceramics would not reach temperatures and stresses as seen in the prototype. Once validated, the latest design was expanded to the originally proposed four sample MCF, manufactured and tested with the water supply. Each unit is located inside a section of the block which makes up the furnace body (Fig. 4).

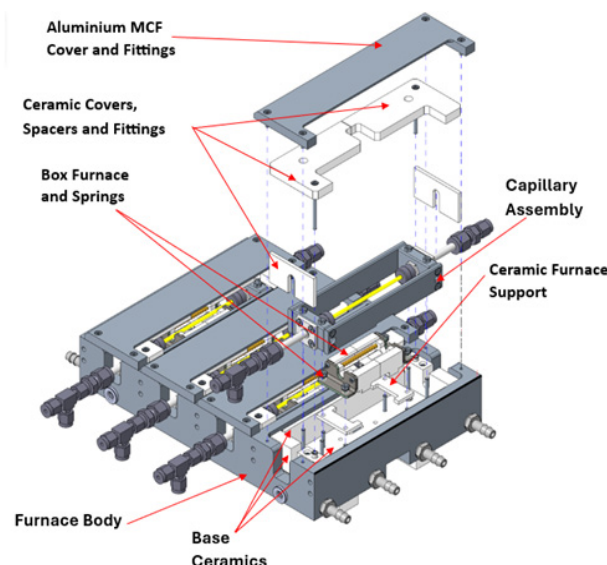


Figure 4: MCF – ‘Box-furnace’ unit and capillary loading process.

EXPERIMENTAL RESULTS

The MCF was tested against the key deliverables by confirming that all samples could be raised from room temperature to the required 600 °C in order to replicate the experimental conditions. Again, this was done by steadily increasing the voltage on the four independent power supplies, with thermocouples near the sample position. In this instance, all samples reached 600 °C, whilst the ‘policeman’ thermocouples on the body maintained an acceptable temperature of lower than 30 °C (Fig. 5). The ceramics which cracked at the prototype testing were redesigned, replaced and also managed to be kept intact.

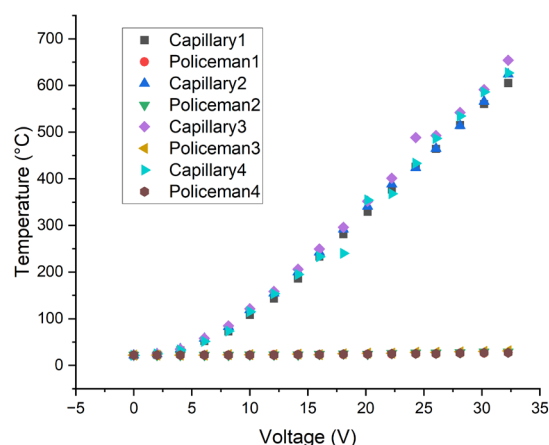


Figure 5: Results for all four capillaries at 600 °C.

Secondly, another experiment was performed wherein the temperature of two samples was raised to 600 °C, but the other two capillaries had no power supplied and were at room temperature. This would test that both the samples maintained stability at their set temperature, and also that the neighbouring samples were unaffected. The test showed that heating only the first and third sample

capillaries in the MCF to 600 °C has a negligible effect on neighbouring capillaries, allowing for simultaneous independent catalysis reactions (see Tables 1 and 2).

Table 1: Sample Temperatures When First and Third Capillaries Temperature Set Up to 600 °C.

T1 (Set) (°C)	T3 (Set) (°C)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)
100	100	100±5	22	100±5	26.3
200	200	200±1	22	200±1	24.7
300	300	300±1	22	300±1	24.0
400	400	400±0.5	22	400±0.5	23.8
500	500	500±0.5	23	500±0.5	23.8
600	600	600±0.5	23	600±0.5	24.2

Table 2: MCF Body Temperature at Policeman Thermocouples When First and Third Capillaries Temperature Set Up to 600 °C.

T1 (Set) (°C)	T3 (Set) (°C)	P1 (°C)	P2 (°C)	P3 (°C)	P4 (°C)
100	100	23	23	22	22
200	200	23	23	23	22
300	300	23	24	23	23
400	400	24	24	23	23
500	500	24	25	23	24
600	600	25	25	24	24

This has prepared the furnace for beamline commissioning where simultaneous reactions will occur across all samples, and motion stages are to be used to remotely move the furnace and these samples into the path of the beam.

CONCLUSION

The MCF fulfilled its key deliverable of allowing up to four samples to have concurrent reaction and reach the 600 °C target temperature. It was also able to maintain temperature stability and keep samples thermally isolated from those in adjacent capillaries. This design could potentially be used as a basis to develop future projects where multiple simultaneous sample reactions are required, and motion stages to be used on the beamline experiment tables to move into the beam path.